

Ultrafiltration

Final Lab Report 5
Unit Operations Lab 2
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Group 2
Anna Borkowska
Aref Wahdan
Chris Kozlowski
Alex Kubinski
Eric Adupong

Executive Summary

Two experiments were performed to measure the efficiency of the ultrafiltration system. The first experiment was to add 20L of water at 104°F to the membrane system to observe the permeate flux change with respect to pressure. With a constant inlet pressure of 25 psi and varied outlet pressures of 5, 10, 15, and 20 psig; the flux (L/m^2) obtained was 5.88e^{-6} , 7.32e^{-6} , 8.68e^{-6} , and 9.62e^{-6} ; respectively. The results show that increasing with outlet pressure will also increase the flux. The second experiment was to add one gallon of 1% lowfat milk to deionized water for an 80L solution. The permeate will then be removed from the solution via separation using several tested recoveries of 10, 25, and 40% at 5 pressures. The results display that increasing the outlet pressure with lower recovery will have an increase of flux.

Introduction

Ultrafiltration is a type of membrane filtration that uses pressure or concentration gradients to separate a solution through a semipermeable membrane.¹ A solution is pumped across a membrane that is permeable to the solvent but retains all the solutes that have a molecular weight above the molecular weight cutoff of the membrane.² Suspended solids and solutes of high molecular weight are retained in the retentate, while water and low molecular weight solutes pass through the membrane in the permeate.¹ This process can be used to purify a solvent by removing solute from a solution, increase the concentration of a solute by removing solvent from solution, and separating a solute A from solute B by using a membrane permeable to either A or B.² In the chemical engineering industry, ultrafiltration is used for cheese manufacturing, the removal of particulates and macromolecules from water for consumption, and enzyme recovery.³

The purpose of this experiment is to measure the efficiency of the ultrafiltration system that is used to separate a milk and water solution. This is significant because different system pressures will result in a different flux and therefore affect the efficiency. This experiment shows what the actual effect of pressure is on the flux and the efficiency. The apparatus consist of two pumps, multiple valves, pressure gauges, a membrane column, a permeate tank, and a solution tank. The membrane column is a hollow fiber module that is connected to the tanks. The feed pump delivers a constant head of pressure and valves on the inlet and outlet to the module are adjusted to fix the inlet and outlet pressures.² For the first part of the experiment, a water flux test was done. This was done by running 20 L of 105°F deionized water through the system with an inlet pressure of 25 psi and an outlet pressure of 5, 10, 15, and 20 psi. A sample was collected in a large graduated cylinder for 30 seconds at each pressure. For the next part of the experiment, the process tank was filled with 80 liters of 105°F deionized and 1 gallon of 1% lowfat milk. The system was then run with an inlet pressure of 25 psi and an outlet pressure of 5 psi until 10% recovery was reached. Once 10% recovery was reached the permeate line was returned to the process tank and permeate flux measurements were taken. The outlet pressures were then changed to 15, 10, and 20 psi for the same percent recovery and permeate flux measurements were taken 2 minutes after the outlet pressures were changed. This procedure was then repeated for 25% and 40% recovery.

Theory

In ultrafiltration, a solution is pumped across a membrane that is permeable to the solvent but does not allow solutes of a molecular weight above the molecular weight cutoff pass through the membrane. Essentially, separation occurs based on size exclusion or particle capture.⁴ This is different from membrane gas separation which uses absorption and different rates of diffusion to separate.⁴ In this experiment, the ultrafilter used was a hollow fiber membrane that consisted of a bundle of small hollow tubes.² A milk and water solution was passed through the ultrafilter to separate the milk (concentrate) from the water (permeate). The optimal operating pressure of the system was determined by graphing the Concentration Factor (CF) versus the flux.² The concentration factor is defined as:²

$$CF = \frac{V_{original}}{V_{concentrate}} = \frac{V_{original}}{V_{original} - V_{permeate}}$$

Where V is either the original, permeate, or concentrate volumetric flowrate. Another equation for concentration factor is:²

$$CF = \frac{1}{1-R} \text{ where } R = \frac{V_{permeate}}{V_{original}}$$

A plot of permeate flux versus concentration factor was used to determine the pressure differential where the greatest flux is obtained. The permeate flux is the rate of permeate flowing through a unit area of the membrane surface.²

The basic model of ultrafiltration can be expressed the following way:²

$$v = Q_v \Delta P$$

Where Q_v is the membrane permeability for the solvent and v is the volume flux. It can be seen from this expression that the flux is a function of the applied pressure ΔP . The expression also assumes that osmotic pressure effects do not influence the driving force and that the solutes do not permeate through the membrane. Because the solute does not permeate through the surface, it concentrates at the membrane surface and causes concentration polarization. Concentration polarization is important in the ultrafiltration process and can be seen in the following equations:²

$$C_1 v = -D_v \frac{dC_1}{dx} \text{ (mass balance over a thin element near the membrane surface)}$$

$$v = k_c \ln\left(\frac{C_s}{C_1}\right) \text{ (solution to the previous differential equation with certain boundary conditions)}$$

Where C_s is the solute concentration at the surface, k_c is the mass transfer coefficient, D_v is the solute diffusion coefficient, and C_1 is the solute concentration in the bulk of the solution.² From the solved differential equation it can be seen that pressure is not the driving force for the separation, therefore the concentration gradient is the driving force.⁵ If C_s small then k_c/v is greater than 1, and a greater separation is achieved. Therefore, concentration polarization should be kept at a minimum. This can be achieved by increasing the flow rates of the solutions between the membranes.⁵ In this experiment the outlet pressure was changed for the system at different recovery percentages. This pressure change resulted in a change in flux that was measured and graphed. From the data collected the efficiency of the system was determined based on what pressure settings gave the largest flux.

Results

In this experiment, 1% lowfat milk was ran through a filtration membrane system to observe how different pressures and recoveries affected the filtering of the milk. The outlet pressures were varied from 5, 10, 15, and 20 psig at different recoveries of 10%, 25 % and 40%. Prior to performing the milk experiment, water was used in the membrane system to observe the permeate flux changed with respect to pressure. Using a stopwatch and graduated cylinder, permeate was collected for 30 seconds at four different pressures. The effect of pressure and flux can be seen in *Figure 1*.

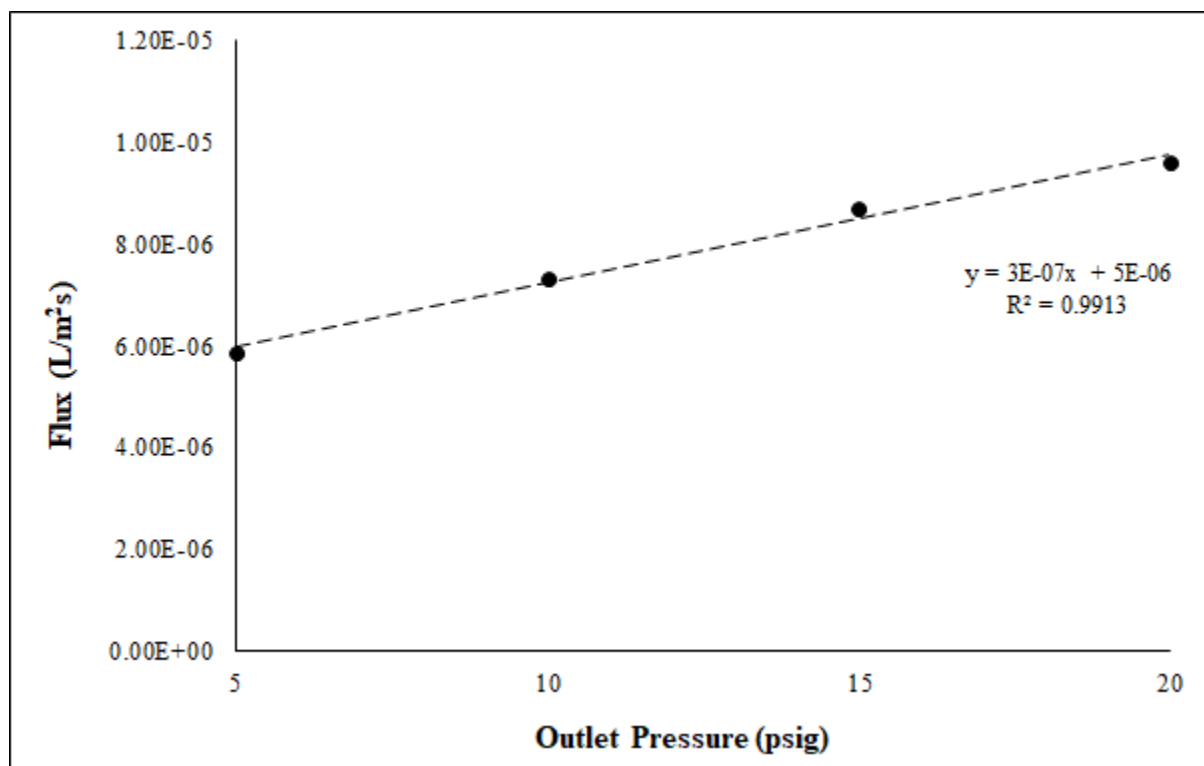


Figure 1: Pure Water Outlet Pressure vs Flux

After performing the water experiment, one gallon of lowfat milk was combined with water at 104 °F. Water was added until the total volume in the tank was 80 L and then stirred with a stirring rod. After circulating the system, 8 L of permeate was removed for 10% recovery. The experiment was then performed by setting the outlet pressure to 5, 10, 15, 20, and then back to 5 psig. At each of these pressures, permeate was collected for the same time in order to calculate the flux for that recovery. This procedure was then performed for 25% and 40% recovery by removing 12 L of permeate for each stage. At 10% recovery and 20 psig, the greatest flux value was obtained. The lowest flux value was obtained at the lowest outlet pressure of 5 psig and 40% recovery. Therefore, the flux increases at higher outlet pressures and lower recovery. *Figures 2 & 3* illustrate how the outlet pressure and pressure drop affect the flux.

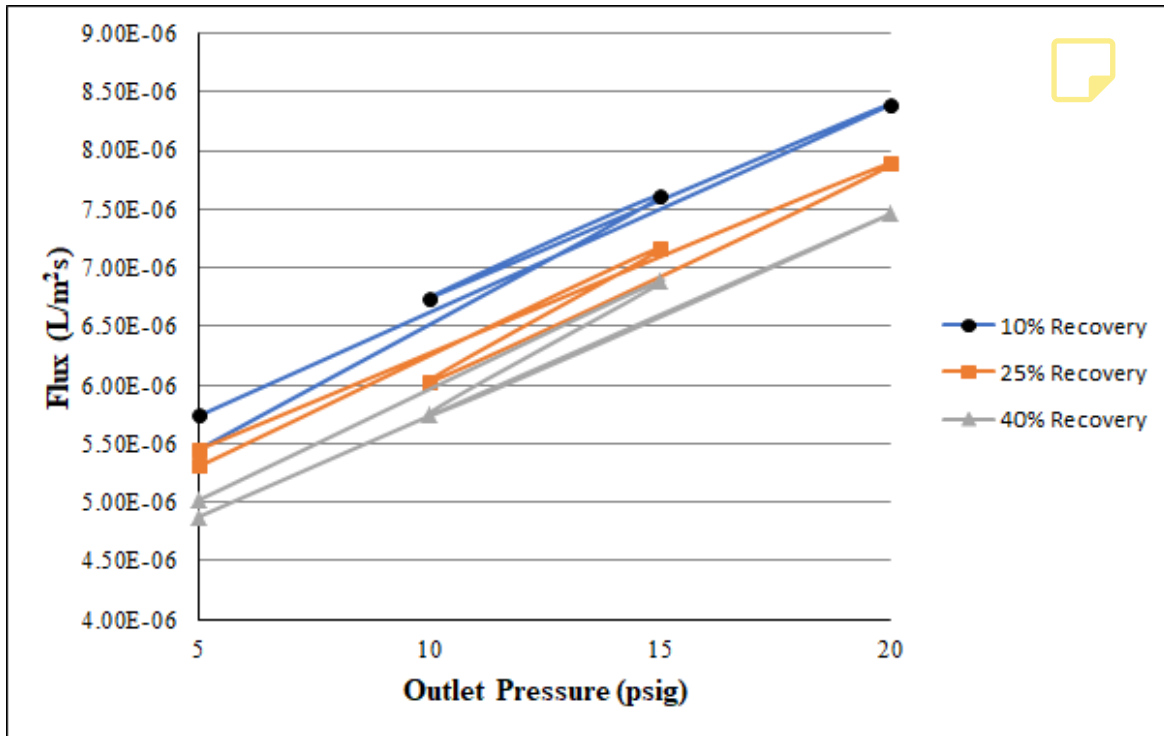


Figure 2: Outlet Pressure vs Flux at Different Recoveries

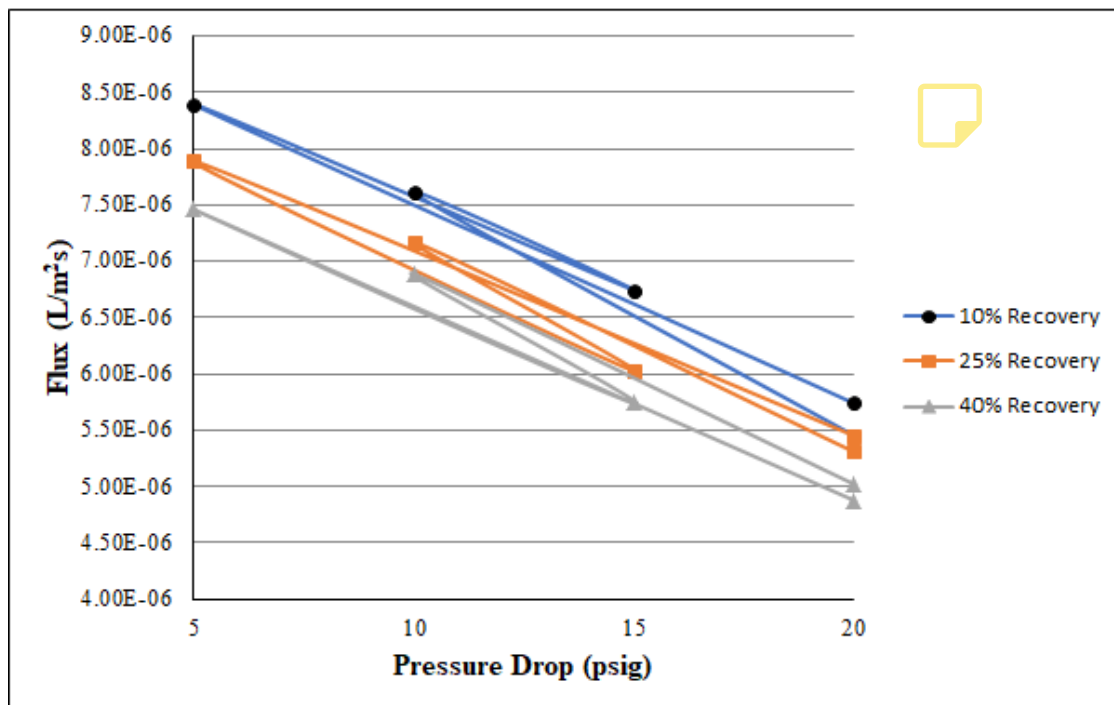


Figure 3: Pressure Drop vs Flux at Different Recoveries

There is also a correlation between the concentration factor and the flux of the system. *Figure 4*, illustrates how the concentration factor affects the flux at different pressure drops. At 5 psig there is the greatest amount of flux. Analyzing the trends of the data we can note that at higher pressure drops, there is an increased flux value.

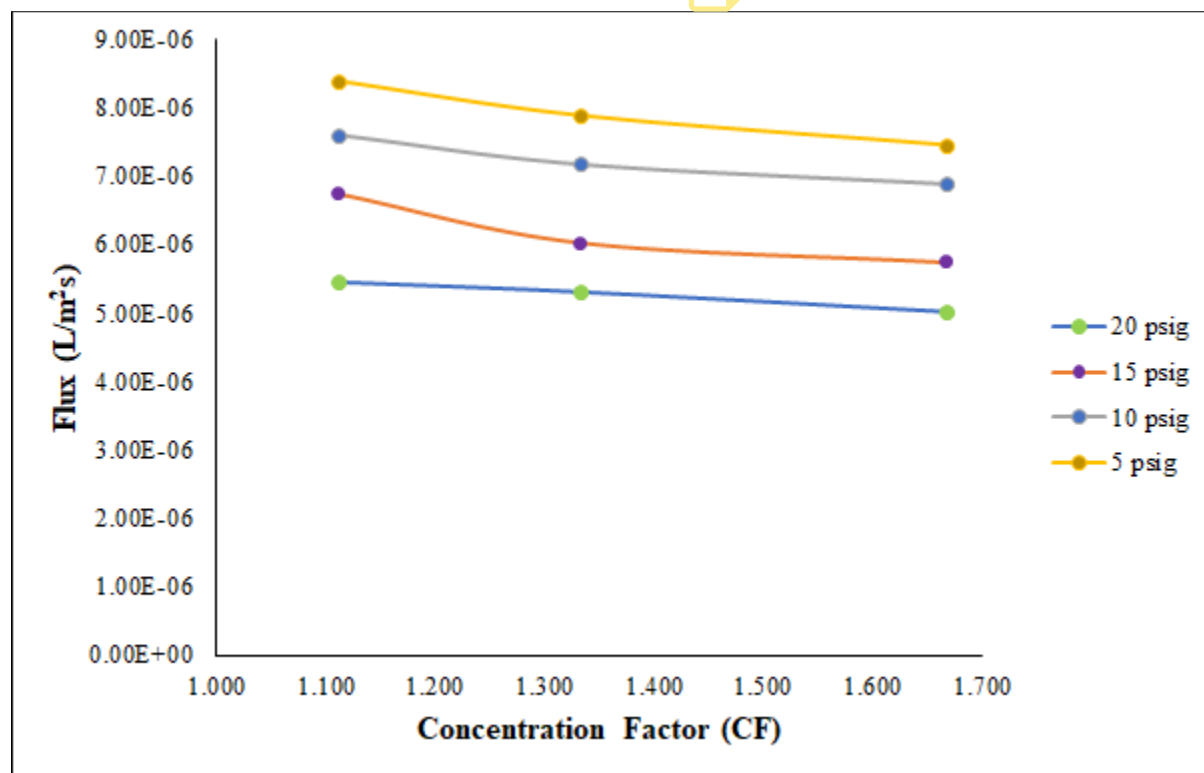


Figure 4 : Concentration Factor vs Flux for different Pressure drops

Discussion

In this experiment, two experiments were performed in order to observe the concepts in the membrane filtration system. The water experiment allowed us to analyze how water would flow through the membrane system. In the pure water experiment, different pressures were used to establish a pressure differential across the column. *Figure 1* illustrates how changing the outlet pressure affects the flux. By increasing the outlet pressure, the flux of the water increases. This is due to the decreased differential pressure across the column. The inlet pressure was kept stable throughout the experiment at 25 psig. The outlet pressure was altered. The flux was greatest when the differential pressure was 5 psig. The R^2 value from the linear trendline was 0.9913 which is close to the expected value of 1.

For the second experiment, 1% lowfat milk was mixed with warm water in order to produce a 80L solution. This mixture was then circulated through the membrane system. In order to have 10% recovery, 8 L of permeate was removed from the cycle. The outlet pressures were then set to 5, 10, 15, 20, and 5 psig. A sample was collected for 30 seconds to calculate the flux discussed in the results section. Based on the data graphed in *Figure 2*, the lowest recovery and highest outlet pressure yields the greatest flux. Whereas, the highest recovery and lowest outlet pressure, 40% and 5 psig, has the lowest flux.

Membrane permeability involves the membrane acting as a specific filter that will let water flow through while it catches suspended solids. In this experiment, the membrane is selective more of the water rather than the proteins, mostly casein and whey, of the milk. Likewise, experiments carried out at lower recovery, 10%, refer to the removal of 10% of the solution where the permeate removed consists mainly of water. Therefore, the milk proteins are recycled back into the feed which will increase the concentration of the feed. The concentration factor was calculated to determine the ratio of solute in the feed to the solute in the concentrate. It was then plotted with flux obtained at the various outlet pressures as seen in *Figure 4*.

References

- 1.) "Ultrafiltration." <<https://en.wikipedia.org/wiki/Ultrafiltration>>
- 2.) W.L. McCabe, J.C. Smith, P. Harriott, "Unit Operations of Chemical Engineering", 7th ed. McGraw-Hill, 2005.
- 3.) R.W. Baker, B.T. Low. "Membrane Separations" Encyclopedia of Separation Science, 2000.
- 4.) P.C. Wankat, *Separation Process Engineering*, 4th ed. Prentice Hall, 2017.
- 5.) "Concentration polarization." <https://en.wikipedia.org/wiki/Concentration_polarization>

Appendix I: Data Tabulations/Graphs

Table 1: Membrane Properties

Membrane Properties		
Membrane Area	25.0	ft ²
	2.32	m ²

Table 2: Water Experiment Flux and Pressure changes

Water Experiment						
Outlet Pressure (psig)	Time (s)	dP	Volume (mL)	Flow Rate (mL/s)	Flow Rate (L/s)	Flux (L/s m ²)
5	30	20	410	13.7	0.0137	5.88E-06
10	30	15	510	17.0	0.0170	7.32E-06
15	30	10	605	20.2	0.0202	8.68E-06
20	30	5	670	22.3	0.0223	9.62E-06

Table 3: Milk Experiment 10 % Recovery

Milk Experiment (Recovery 10%)						
Outlet Pressure (psig)	Time (s)	dP	Volume (mL)	Flow Rate (mL/s)	Flow Rate (L/s)	Flux (m/s)
5	30	20	380	12.7	0.0127	5.45E-06
15	30	10	530	17.7	0.0177	7.61E-06
10	30	15	470	15.7	0.0157	6.75E-06
20	30	5	585	19.5	0.0195	8.40E-06
5	30	20	400	13.3	0.0133	5.74E-06

Table 4: Milk Experiment 25 % Recovery

Milk Experiment (Recovery 25%)						
Outlet Pressure (psig)	Time (s)	dP	Volume (mL)	Flow Rate (mL/s)	Flow Rate (L/s)	Flux (m/s)
5	30	20	370	12.3	0.0123	5.31E-06
15	30	10	500	16.7	0.0167	7.18E-06
10	30	15	420	14.0	0.0140	6.03E-06
20	30	5	550	18.3	0.0183	7.89E-06

5	30	20	380	12.7	0.0127	5.45E-06
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Table 5: Milk Experiment 40 % Recovery

Milk Experiment (Recovery 40%)						
Outlet Pressure (psig)	Time (s)	dP	Volume (mL)	Flow Rate (mL/s)	Flow Rate (L/s)	Flux (m/s)
5	30	20	350	11.7	0.0117	5.02E-06
15	30	10	480	16.0	0.0160	6.89E-06
10	30	15	400	13.3	0.0133	5.74E-06
20	30	5	520	17.3	0.0173	7.46E-06
5	30	20	340	11.3	0.0113	4.88E-06

Table 6: Concentration Factor and Pressure

	Pressure Drop			
Concentration Factor	5 PSIG	10 PSIG	15 PSIG	20 PSIG
1.111	8.40E-06	7.61E-06	6.75E-06	5.45E-06
1.333	7.89E-06	7.18E-06	6.03E-06	5.31E-06
1.667	7.46E-06	6.89E-06	5.74E-06	5.02E-06

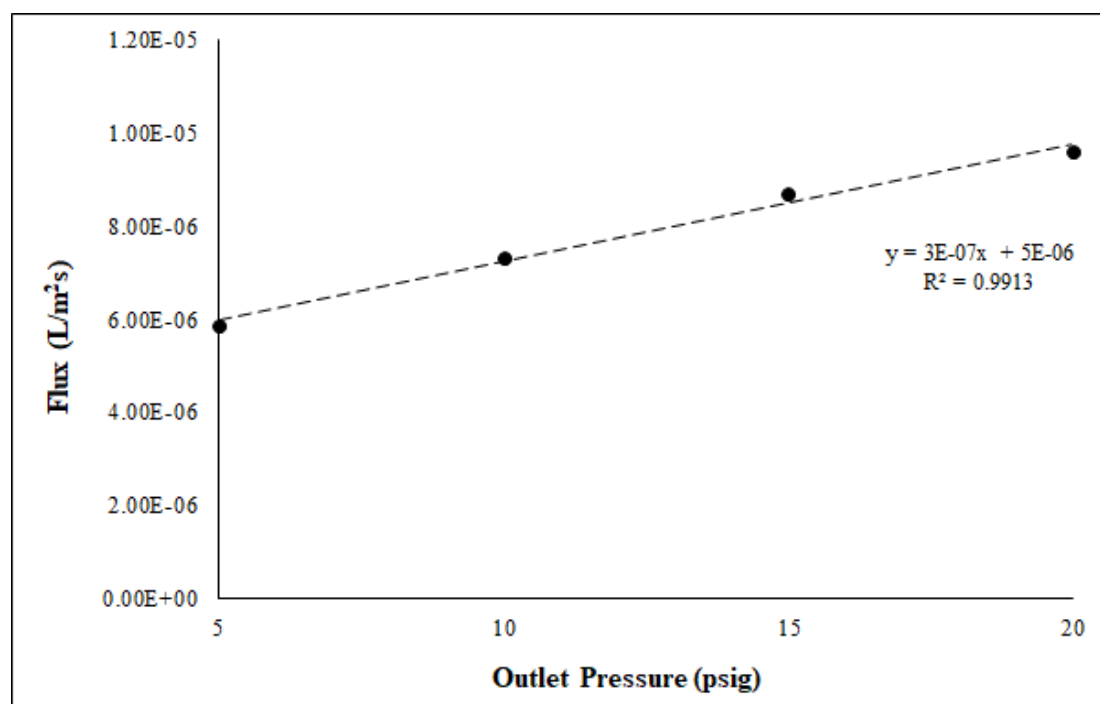


Figure 1: Pure Water Outlet Pressure vs Flux

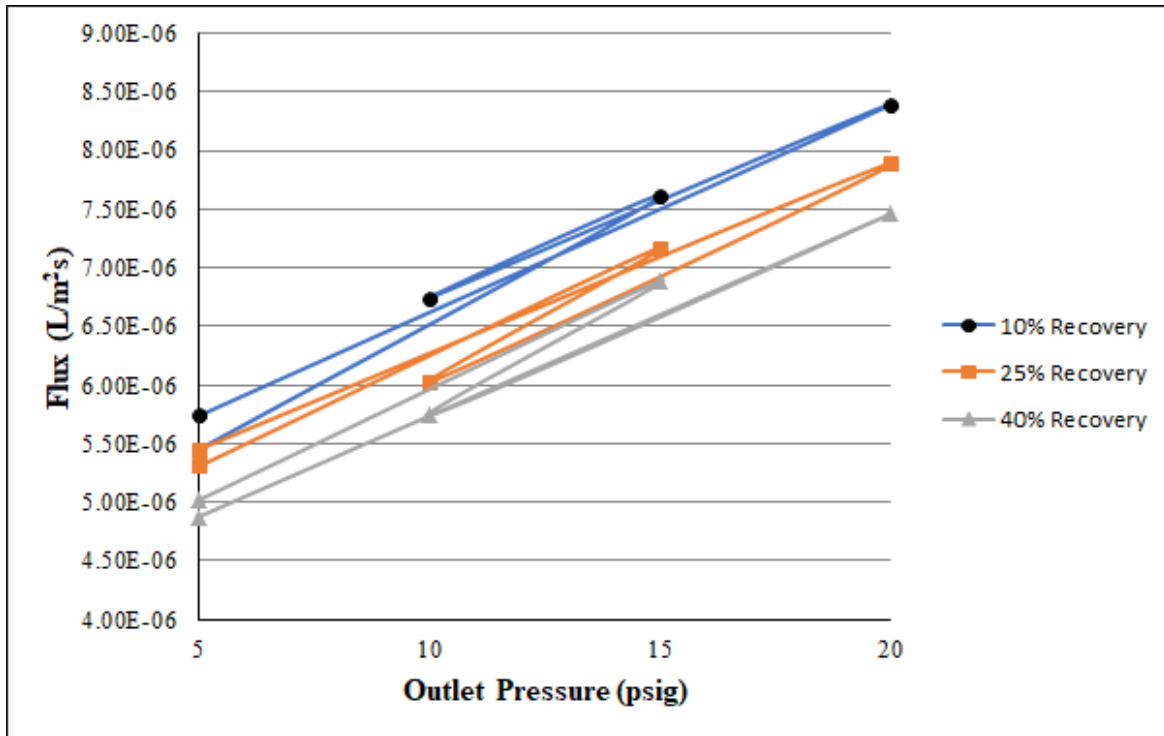


Figure 2: Outlet Pressure vs Flux at Different Recoveries

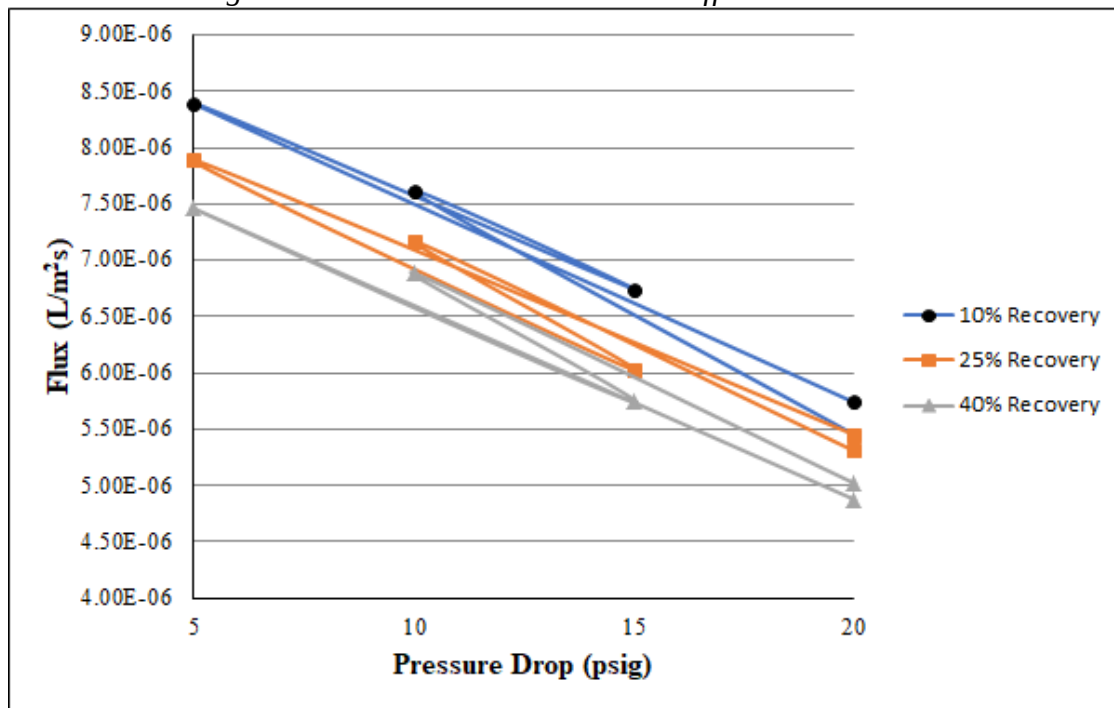


Figure 3: Pressure Drop vs Flux at Different Recoveries

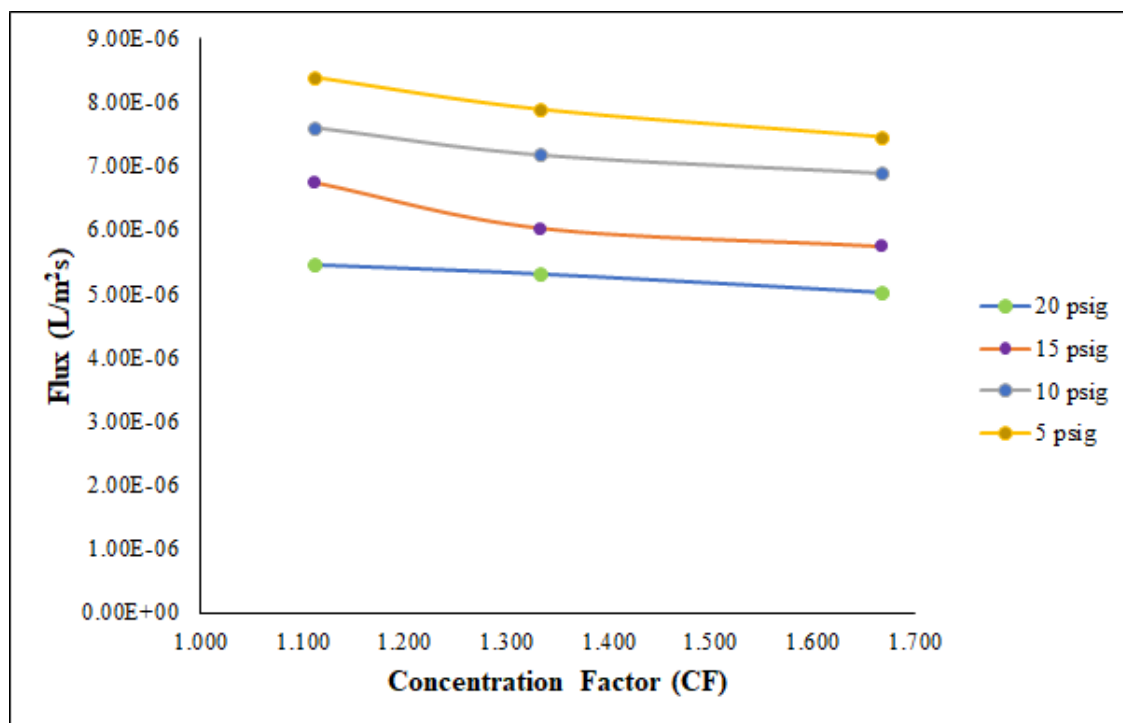


Figure 4 : Concentration Factor vs Flux for different Pressures

Appendix II: Error Analysis

There are many pieces of measurement equipment involved in this experiment that may have been the source of error. There were two pressure gauges, one thermocouple, the main tank, the recovery tank, and one graduated cylinder. All of these instruments have some amount of random error associated with them. With the pressure gauges and thermocouple it is minimal and the same throughout the experiment. The main source of random error is the lines for measuring the amount of solution in the main tank and recovery tank. This is an issue when filling the main to the starting volume of 80L. It is also an issue when collecting permeate to achieve the varying recovery amounts, such as when collecting 8L to achieve a 10% recovery.

There is some operator error involved in reading the pressure gauges and measuring the permeate in a graduated cylinder. To combat this the same person would be the one to take readings from their assigned instrument so this would remain consistent. There is assumed to not be any operator error for the thermocouple as it was a digital reading for regulating the temperature of the feed. There is also some operator error in the collection of the permeate. For the collection of permeate, one person would be using the timer and another operator would have to close the appropriate valves. This may have lead to a time delay between the time measurement and the sample collection. This makes the final calculation of flux to not be completely accurate for the amount collected vs the amount of time it was collected.

Another source of operator error is in the pre-lab cleaning procedure of the apparatus. The afternoon lab group performed steps 3-5 of the cleaning procedure. While, the morning lab group performed steps 1-3 for their pre-lab. The morning group then also performed step 6 of the procedure the morning of the actual lab. Each group has to assume that the other did their

perspective portion of the cleaning procedure correctly. Improper execution of this procedure can result in a dirty membrane that would not achieve the proper amount of permeate flux.

Appendix III: Sample Calculations

$$\text{Membrane area} = 25.0 \text{ ft}^2 * 0.092903 = 2.32 \text{ m}^2$$

1. Pressure drop

$$\begin{aligned}\Delta P \text{ (psig)} &= \text{Inlet pressure} - \text{Outlet pressure} \\ \Delta P \text{ (psig)} &= 25 - 5 = 20 \text{ (psig)}\end{aligned}$$

2. Percent Recovery:

$$\begin{aligned}\text{Percent Recovery} &= \frac{\text{Permeate Removed [L]}}{\text{Original Volume [L]}} * 100 = \frac{32 \text{ L}}{80.0 \text{ L}} * 100 \\ &= 40\% \text{ Recovery}\end{aligned}$$

3. Permeate flowrate (L/s)

$$\begin{aligned}\text{Permeate Flowrate (L/s)} &= \frac{\text{Permeate Collected Volume (l)}}{\text{Time Permeate Collected (s)}} \\ &= \frac{0.350 \text{ L}}{30 \text{ s}} = 0.117 \text{ L/s}\end{aligned}$$

4. Permeate Flux (m/s)

$$\begin{aligned}\text{Permeate Flux (m/s)} &= \frac{\text{Permeate flowrate (l/s)}}{\text{Membrane area (m}^2\text{)}} \\ &= \frac{0.0117 \text{ L/s}}{2.32 \text{ m}^2} * 0.001 = 5.02E - 6 \text{ (m/s)}\end{aligned}$$

5. Concentration factor (CF)

$$CF = \frac{1}{1 - \text{Recovery}} = \frac{1}{1 - 0.4} = 1.667$$

Appendix IV: Individual Team Contributions

NAME: Anna Borkowska

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Was present for both lab periods
Pre-Lab Editing	1	
Experimental Objective (Prelab)		
Apparatus (prelab)	1	Created figure with numbers and table with descriptions
Materials & Supplies (prelab)		
Procedure (prelab)	1	Wrote the entire procedure
Project Safety Evaluation (prelab)		
Final Lab Editing	1	
Executive Summary (final lab)		
Introduction (Final lab)	3	Wrote this entire portion
Theory (Final Lab)	3	Wrote this entire portion
Results (final lab)		
Discussion (final lab)		

References (final and/or prelab)	1	Added five sources and cited throughout the report
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation		
Total	18	

NAME: Aref Wahdan

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present both lab periods
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)	1	Did half the apparatus
Materials & Supplies (prelab)	1	Did half the materials
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	Edited final

Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	4	Wrote the entire procedure
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1	Did the graphs
Error Analysis (final lab)	1	Wrote the entire procedure
Sample Calculations (final Lab)		
Power Point Presentation		
Total	16	

NAME: Christian Kozlowski

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present both days
Pre-Lab Editing		

Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)	1	Did some of the materials
Procedure (prelab)		
Project Safety Evaluation (prelab)	1	Did this part
Final Lab Editing	2	Edited lab
Executive Summary (final lab)	1	Did the entire summary
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	2	
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1	Did tables
Error Analysis (final lab)		
Sample Calculations (final Lab)	1	Did the calculations

Power Point Presentation	1	Did the procedure
Total	17	

NAME: Alex Kubinski

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present for both lab periods
Pre-Lab Editing		
Experimental Objective (Prelab)	1	
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	2	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	1	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	1	

Discussion (final lab)	1	
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1	
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	
Total	16	

NAME: Eric Adupong

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present for both lab periods
Pre-Lab Editing	1	
Experimental Objective (Prelab)	1	
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	1	

Project Safety Evaluation (prelab)	1	
Final Lab Editing	1	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	1	
Discussion (final lab)	1	
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	2	
Total	16	